

Prove that $\lfloor \alpha n \rfloor + \lceil (1 - \alpha)n \rceil = n$ for any integer n and real number α in the range $0 \leq \alpha \leq 1$.

3.3-3

Use equation (3.14) or other means to show that $(n + o(n))^k = \Theta(n^k)$ for any real constant k . Conclude that $\lceil n \rceil^k = \Theta(n^k)$ and $\lfloor n \rfloor^k = \Theta(n^k)$.

3.3-4

Prove the following:

a. Equation (3.21).

b. Equations (3.26)–(3.28).

c. $\lg(\Theta(n)) = \Theta(\lg n)$.

★ 3.3-5

Is the function $\lceil \lg n \rceil!$ polynomially bounded? Is the function $\lceil \lg \lg n \rceil!$ polynomially bounded?

★ 3.3-6

Which is asymptotically larger: $\lg(\lg^* n)$ or $\lg^*(\lg n)$?

3.3-7

Show that the golden ratio ϕ and its conjugate $\hat{\phi}$ both satisfy the equation $x^2 = x + 1$.

3.3-8

Prove by induction that the i th Fibonacci number satisfies the equation

$$F_i = (\phi^i - \hat{\phi}^i) / \sqrt{5},$$

where ϕ is the golden ratio and $\hat{\phi}$ is its conjugate.

3.3-9

Show that $k \lg k = \Theta(n)$ implies $k = \Theta(n/\lg n)$.

Problems

3-1 Asymptotic behavior of polynomials

Let

$$p(n) = \sum_{i=0}^d a_i n^i,$$

where $a_d > 0$, be a degree- d polynomial in n , and let k be a constant. Use the definitions of the asymptotic notations to prove the following properties.

a. If $k \geq d$, then $p(n) = O(n^k)$.

b. If $k \leq d$, then $p(n) = \Omega(n^k)$.

c. If $k = d$, then $p(n) = \Theta(n^k)$.

d. If $k > d$, then $p(n) = o(n^k)$.

e. If $k < d$, then $p(n) = \omega(n^k)$.

3-2 Relative asymptotic growths

Indicate, for each pair of expressions (A , B) in the table below whether A is O , o , Ω , ω , or Θ of B . Assume that $k \geq 1$, $\epsilon > 0$, and $c > 1$ are constants. Write your answer in the form of the table with “yes” or “no” written in each box.

	A	B	O	o	Ω	ω	Θ
a.	$\lg^k n$	n^ϵ					
b.	n^k	c^n					
c.	$\sqrt{n}$	$n^{\sin n}$					
d.	2^n	$2^{n/2}$					
e.	$n^{\lg c}$	$c^{\lg n}$					
f.	$\lg(n!)$	$\lg(n^n)$					

3-3 Ordering by asymptotic growth rates

- a.** Rank the following functions by order of growth. That is, find an arrangement $g_1, g_2, \dots, g_{30}$ of the functions satisfying $g_1 = \Omega(g_2)$, $g_2 = \Omega(g_3)$, $\dots$, $g_{29} = \Omega(g_{30})$. Partition your list into equivalence classes such that functions $f(n)$ and $g(n)$ belong to the same class if and only if $f(n) = \Theta(g(n))$.

$$\begin{array}{ccccccc} \lg(\lg^* n) & 2^{\lg^* n} & (\sqrt{2})^{\lg n} & n^2 & n! & (\lg n)! & \\ (3/2)^n & n^3 & \lg^2 n & \lg(n!) & 2^{2^n} & n^{1/\lg n} & \\ \ln \ln n & \lg^* n & n \cdot 2^n & n^{\lg \lg n} & \ln n & 1 & \\ 2^{\lg n} & (\lg n)^{\lg n} & e^n & 4^{\lg n} & (n+1)! & \sqrt{\lg n} & \\ \lg^*(\lg n) & 2^{\sqrt{2 \lg n}} & n & 2^n & n \lg n & 2^{2^{n+1}} & \end{array}$$

- b.** Give an example of a single nonnegative function $f(n)$ such that for all functions $g_i(n)$ in part (a), $f(n)$ is neither $O(g_i(n))$ nor $\Omega(g_i(n))$.

3-4 Asymptotic notation properties

Let $f(n)$ and $g(n)$ be asymptotically positive functions. Prove or disprove each of the following conjectures.

- $f(n) = O(g(n))$ implies $g(n) = O(f(n))$.
- $f(n) + g(n) = \Theta(\min \{f(n), g(n)\})$.
- $f(n) = O(g(n))$ implies $\lg f(n) = O(\lg g(n))$, where $\lg g(n) \geq 1$ and $f(n) \geq 1$ for all sufficiently large n .
- $f(n) = O(g(n))$ implies $2^{f(n)} = O(2^{g(n)})$.
- $f(n) = O((f(n))^2)$.
- $f(n) = O(g(n))$ implies $g(n) = \Omega(f(n))$.

g. $f(n) = \Theta(f(n/2))$.

h. $f(n) + o(f(n)) = \Theta(f(n))$.

3-5 *Manipulating asymptotic notation*

Let $f(n)$ and $g(n)$ be asymptotically positive functions. Prove the following identities:

a. $\Theta(\Theta(f(n))) = \Theta(f(n))$.

b. $\Theta(f(n)) + O(f(n)) = \Theta(f(n))$.

c. $\Theta(f(n)) + \Theta(g(n)) = \Theta(f(n) + g(n))$.

d. $\Theta(f(n)) \cdot \Theta(g(n)) = \Theta(f(n) \cdot g(n))$.

e. Argue that for any real constants $a_1, b_1 > 0$ and integer constants k_1, k_2 , the following asymptotic bound holds:

$$(a_1 n)^{k_1} \lg^{k_2}(a_2 n) = \Theta(n^{k_1} \lg^{k_2} n).$$

★ **f.** Prove that for $S \subseteq \mathbb{Z}$, we have

$$\sum_{k \in S} \Theta(f(k)) = \Theta\left(\sum_{k \in S} f(k)\right),$$

assuming that both sums converge.

★ **g.** Show that for $S \subseteq \mathbb{Z}$, the following asymptotic bound does not necessarily hold, even assuming that both products converge, by giving a counterexample:

$$\prod_{k \in S} \Theta(f(k)) = \Theta\left(\prod_{k \in S} f(k)\right).$$

3-6 *Variations on O and Ω*

Some authors define Ω -notation in a slightly different way than this textbook does. We'll use the nomenclature $\tilde{\Omega}$ (read “omega infinity”) for this alternative definition. We say that $f(n) = \tilde{\Omega}(g(n))$ if there exists a

positive constant c such that $f(n) \geq cg(n) \geq 0$ for infinitely many integers n .

- a.** Show that for any two asymptotically nonnegative functions $f(n)$ and $g(n)$, we have $f(n) = O(g(n))$ or $f(n) = \tilde{\Omega}(g(n))$ (or both).
- b.** Show that there exist two asymptotically nonnegative functions $f(n)$ and $g(n)$ for which neither $f(n) = O(g(n))$ nor $f(n) = \Omega(g(n))$ holds.
- c.** Describe the potential advantages and disadvantages of using $\tilde{\Omega}$ -notation instead of Ω -notation to characterize the running times of programs.

Some authors also define O in a slightly different manner. We'll use O' for the alternative definition: $f(n) = O'(g(n))$ if and only if $|f(n)| = O(g(n))$.

- d.** What happens to each direction of the “if and only if” in Theorem 3.1 on page 56 if we substitute O' for O but still use Ω ?

Some authors define $\tilde{O}$ (read “soft-oh”) to mean O with logarithmic factors ignored:

$$\tilde{O}(g(n)) = \{f(n) : \text{there exist positive constants } c, k, \text{ and } n_0 \text{ such that } 0 \leq f(n) \leq cg(n) \lg^k(n) \text{ for all } n \geq n_0\}.$$

- e.** Define $\tilde{\Omega}$ and $\tilde{\Theta}$ in a similar manner. Prove the corresponding analog to Theorem 3.1.

3-7 Iterated functions

We can apply the iteration operator $*$ used in the $\lg^*$ function to any monotonically increasing function $f(n)$ over the reals. For a given constant $c \in \mathbb{R}$, we define the iterated function f_c^* by

$$f_c^*(n) = \min \{i \geq 0 : f^{(i)}(n) \leq c\},$$

which need not be well defined in all cases. In other words, the quantity $f_c^*(n)$ is the minimum number of iterated applications of the function f required to reduce its argument down to c or less.

For each of the functions $f(n)$ and constants c in the table below, give as tight a bound as possible on $f_c^*(n)$. If there is no i such that $f^{(i)}(n) \leq c$, write “undefined” as your answer.

	$f(n)$	c	$f_c^*(n)$
a.	$n - 1$	0	
b.	$\lg n$	1	
c.	$n/2$	1	
d.	$n/2$	2	
e.	$\sqrt{n}$	2	
f.	$\sqrt{n}$	1	
g.	$n^{1/3}$	2	

Chapter notes

Knuth [259] traces the origin of the O -notation to a number-theory text by P. Bachmann in 1892. The o -notation was invented by E. Landau in 1909 for his discussion of the distribution of prime numbers. The Ω and Θ notations were advocated by Knuth [265] to correct the popular, but technically sloppy, practice in the literature of using O -notation for both upper and lower bounds. As noted earlier in this chapter, many people continue to use the O -notation where the Θ -notation is more technically precise. The soft-oh notation $\tilde{O}$ in Problem 3-6 was introduced by Babai, Luks, and Seress [31], although it was originally written as $O\sim$. Some authors now define $\tilde{O}(g(n))$ as ignoring factors that are logarithmic in $g(n)$, rather than in n . With this definition, we can say that $n2^n = \tilde{O}(2^n)$, but with the definition in Problem 3-6, this statement is not true. Further discussion of the history and development of asymptotic notations appears in works by Knuth [259, 265] and Brassard and Bratley [70].

Not all authors define the asymptotic notations in the same way, although the various definitions agree in most common situations. Some of the alternative definitions encompass functions that are not asymptotically nonnegative, as long as their absolute values are appropriately bounded.

Equation (3.29) is due to Robbins [381]. Other properties of elementary mathematical functions can be found in any good mathematical reference, such as Abramowitz and Stegun [1] or Zwillinger [468], or in a calculus book, such as Apostol [19] or Thomas et al. [433]. Knuth [259] and Graham, Knuth, and Patashnik [199] contain a wealth of material on discrete mathematics as used in computer science.

¹ Within set notation, a colon means “such that.”